

Urban Mobility & Bike Sharing Analytics Dashboard Report

Behavioral, Operational & Geospatial Analysis of BIXI Bike Sharing System

1. Objective

This project focuses on analyzing urban mobility and bike-sharing usage patterns using a Data Studio built on BIXI bike-sharing system data.

The dashboard provides behavioral, operational, and geospatial insights into trip activity, rider segmentation, ride duration, station usage, and peak demand periods.

The report demonstrates how data visualization and business intelligence techniques can be used to transform raw trip records into actionable mobility insights for city planners, transport analysts, and mobility service providers.

2. Dataset Source and Overview

The dataset used for this project was taken from the publicly available BIXI Montréal Bike Sharing Trips (2021) dataset, which contains real-world urban mobility and bike-sharing trip records from Montréal's public bike-sharing network.

Dataset Details:

- Dataset Name: BIXI Montréal Bike Sharing Trips (2021)
- Domain: Urban Mobility / Transportation Analytics
- Geographic Scope: Montréal, Canada
- Data Type: Public Bike Sharing Trip Records
- Source Type: Open Public Transportation Dataset

Official Sources:

- [BIXI Open Data Portal](#)
- [Official BIXI Website](#)

The dataset includes:

- Trip duration
- Rider type (Member/Casual)
- Start station
- Destination station
- Ride timestamps
- Ride categories
- Station-level mobility activity

The BIXI open-data platform provides historical and operational mobility datasets that support urban transportation research, mobility optimization, and behavioral analytics.

Domain: Urban Mobility / Transportation Analytics

Primary Use Case: Behavioral and Operational Analysis

Visualization Tool: Google Data Studio / Looker Studio

Analysis Scope:

- Rider behavior analysis
- Trip distribution analysis
- Station activity analysis
- Peak hour analysis
- Geospatial mobility analysis
- Ride category analysis

3. Business Problem Statement

Urban bike-sharing systems generate massive volumes of trip data every day. However, city administrators and mobility operators often struggle to identify:

- Peak congestion hours
- High-demand stations
- Rider behavioral trends
- Operational bottlenecks
- Member vs casual rider patterns
- Spatial demand concentration

This dashboard addresses those challenges by providing an interactive analytical environment that helps stakeholders understand usage behavior, optimize station placement, improve operational efficiency, and support mobility planning decisions.

4. Dashboard Features & Interactive Components

The dashboard includes the following analytical components:

4.1 KPI Cards

- Total Trips
- Member Trips
- Casual Trips
- Average Daily Trips
- Average Ride Duration

4.2 Interactive Filters

- Member Type Filter
- Ride Category Filter

4.3 Visual Analytics

- Weekday Trip Analysis
- Member vs Casual Rider Distribution
- Ride Category Distribution
- Peak Riding Hours
- Top Start Stations
- Top Destination Stations

- Bike Station Activity Map

4.4 Insight Panel

The dashboard contains a dedicated insight panel summarizing important operational findings and rider behavior observations.

5. Data Cleaning & Transformation

The dataset was cleaned and transformed before dashboard development.

Pre-processing activities included:

- Removing duplicate records
- Handling missing values
- Converting timestamps into date-time format
- Creating weekday and hourly derived columns
- Categorizing ride durations into Short, Medium, and Long rides
- Standardizing station names
- Validating trip duration values
- Creating rider segmentation fields

Derived Features:

- Ride Category
- Peak Hour Classification
- Weekday Name
- Hourly Activity Group
- Rider Type Classification

6. Dashboard Analysis & Insights

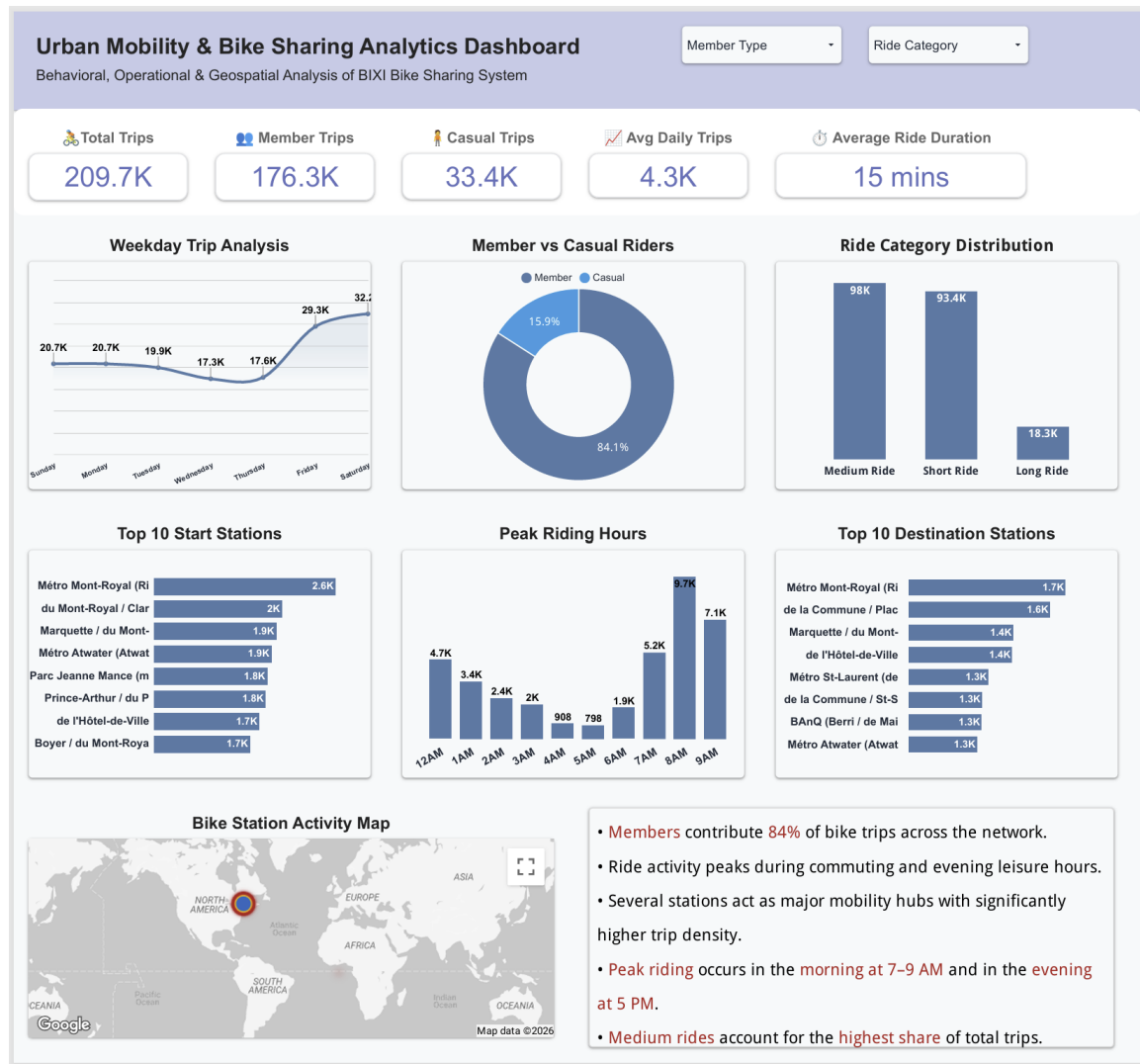


Figure 1: Final Urban Mobility & Bike Sharing Analytics Dashboard

6.1 KPI Summary

The dashboard reports approximately:

- 209.7K Total Trips
- 176.3K Member Trips
- 33.4K Casual Trips
- 4.3K Average Daily Trips
- 15 Minutes Average Ride Duration

Insight:

The platform is heavily dominated by member riders, indicating strong recurring usage behavior.

Business Impact:

This insight supports subscription-focused mobility strategies and long-term customer retention initiatives.

6.2 Weekday Trip Analysis

Analysis:

Weekend activity, especially Saturday, records the highest trip count, while Wednesday and Thursday show comparatively lower activity.

Insight:

Leisure and recreational riding behavior increases significantly during weekends.

Business Recommendation:

Operators can increase bike availability and maintenance readiness before weekend demand surges.

6.3 Member vs Casual Riders

Analysis:

Members contribute approximately 84% of all trips, while casual riders account for roughly 16%.

Insight:

The bike-sharing system is primarily utilized by recurring users rather than occasional riders.

Business Recommendation:

Loyalty programs and premium member services can further strengthen retention and recurring engagement.

6.4 Ride Category Distribution

Analysis:

Medium rides and short rides dominate the overall distribution, while long rides represent a much smaller proportion.

Insight:

Users primarily rely on bikes for short-distance urban commuting and nearby mobility needs.

Business Recommendation:

Station placement should focus on dense urban corridors and commuter routes.

6.5 Peak Riding Hours

Analysis:

Trip activity peaks strongly between 7 AM and 9 AM, with the highest activity around 8 AM.

Insight:

Morning commuting behavior is the strongest demand driver in the system.

Business Recommendation:

Operators should ensure bike availability and redistribution efficiency during commuting windows.

6.6 Top Start & Destination Stations

Analysis:

Stations around Metro Mont-Royal and nearby central locations generate the highest trip activity.

Insight:

Transit-connected stations function as mobility hubs with significantly higher demand.

Business Recommendation:

High-demand stations should receive priority for bike redistribution and operational monitoring.

6.7 Geospatial Activity Map

Analysis:

The activity map visually highlights dense mobility concentration around the operational urban region.

Insight:

Demand concentration is geographically clustered rather than evenly distributed.

Business Recommendation:

Expansion planning should prioritize high-density mobility zones and transit-linked areas.

7. Design & Visualization Best Practices

The dashboard follows modern business intelligence and visualization principles:

- Consistent color palette and typography
- Clear KPI hierarchy

- Interactive filtering capability
- Balanced layout structure
- Combination of categorical, trend, and spatial analysis
- Minimal clutter with business-focused storytelling
- Insight-driven narrative panel

The dashboard structure supports both executive-level overview and operational-level exploration.

8. Descriptive, Diagnostic & Prescriptive Insights

Descriptive Insights:

The dashboard explains how riders use the bike-sharing system across different time periods, rider categories, and locations.

Diagnostic Insights:

The analysis identifies why certain stations experience higher demand and why commuting hours dominate ride activity.

Prescriptive Insights:

Operators can use these findings to:

- Optimize station inventory
- Improve redistribution schedules
- Enhance commuter-focused services
- Design targeted subscription campaigns
- Improve operational efficiency during peak demand periods

9. Conclusion

The Urban Mobility & Bike Sharing Analytics Dashboard successfully transforms raw mobility trip data into meaningful operational and behavioral insights.

The dashboard demonstrates strong analytical storytelling through:

- KPI-driven executive summaries
- Rider segmentation analysis
- Temporal mobility trends
- Geospatial visualization
- Operational station analytics

Overall, the project highlights how modern BI dashboards can support smart mobility planning, operational optimization, and data-driven transportation decision-making.

10. Future Scope

Future enhancements may include:

- Predictive demand forecasting

- Weather-integrated ride analysis
- Real-time mobility monitoring
- Dynamic station occupancy analysis
- Machine learning-based demand prediction
- Route recommendation analysis
- Sustainability and carbon reduction metrics
- Mobile-responsive dashboard integration

11. References

- Google Looker Studio Documentation
- Urban Mobility Analytics Case Studies